Find what is needed to make the forces balance

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Equilibrium / ES: equilibrio / PT: equilíbrio

Acceleration is zero. The forces sum to zero in every direction.

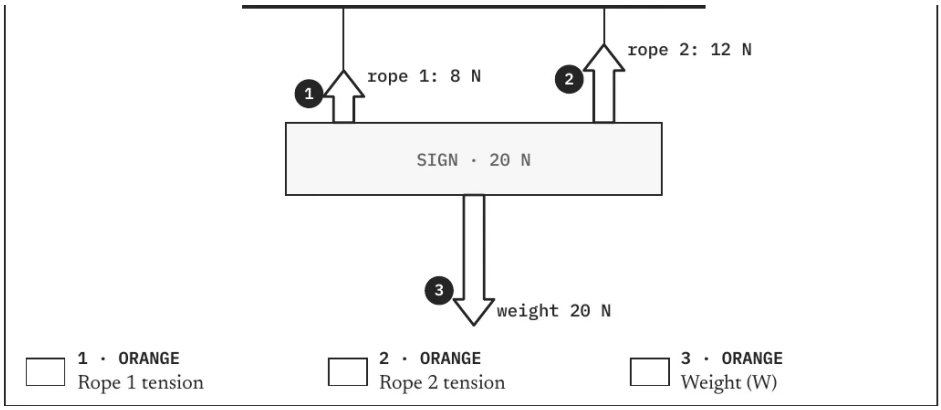
Also on your note page: Static and dynamic equilibrium, Missing force, Instantaneous rest.

B READ THE RELATIONSHIP

An object is in translational equilibrium when its forces add to zero.

Equilibrium requires zero force sum on every axis

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 40 N sign hangs at rest from two vertical ropes. One tension is 15 N. Find the other.

Given weight 40 N rope 1 = 15 N at rest

Rule up = down

Work 15 + T 2 = 40 T 2 = 40 − 15

Answer 25 N up

E SAVE FOR THE FINAL MEETING

If the forces on XJ balanced, its acceleration would be zero and its velocity would stay constant. Analysts check for balance before assuming steady motion.

Read the physics: Equilibrium

SUPPORTED READING / read one part, then point to the evidence Name

01 PICTURE IT

Choose horizontal and vertical axes. Add the forces on each axis. When both sums are zero, acceleration is zero. The object can be at rest or moving steadily.

02 HOW IT WORKS

A hanging sign has downward weight and upward support. If the ropes are vertical, add their tensions. If the ropes are tilted, only the upward parts support the weight.

03 WHAT TO CHECK

Being at rest for an instant is not enough to prove equilibrium. Look for zero net force and unchanged velocity.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 40 N sign hangs at rest from two vertical ropes. One tension is 15 N. Find the other. 25 N up.

05 CONNECT TO XJ

If the forces on XJ balanced, its acceleration would be zero and its velocity would stay constant. Analysts check for balance before assuming steady motion.

SAY IT BEFORE YOU WRITE

Can an object in equilibrium be moving? The ___ forces balance. Its acceleration is ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Equilibrium

ES: equilibrio / PT: equilíbrio

Meaning in my words	How can you tell?
How would you explain equilibrium to a friend? In my words, this means ___.	A cart rolls straight at steady speed. What must the horizontal forces add up to? I can tell because ___.
Example or sketch	**Non-example and why**
Sketch the horizontal forces on a cart with a 3 N pull right and 3 N friction left. My example shows ___.	A tossed ball is briefly still at the top. Is it in equilibrium there? I would change ___ because ___.

Static and dynamic equilibrium

ES: equilibrio estático y dinámico / PT: equilíbrio estático e dinâmico

Meaning in my words	How can you tell?
What do static and dynamic equilibrium have in common? How do they differ? In my words, this means ___.	A parked cart and a cart rolling straight at steady speed both have zero net force. How do they differ? I can tell because ___.
Example or sketch	**Non-example and why**
Draw equal-time dots for each cart. My example shows ___.	A bike turns at steady speed. Would that be dynamic equilibrium? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Missing force

ES: fuerza desconocida / PT: força desconhecida

Meaning in my words	How can you tell?
How would you explain missing force to a friend? In my words, this means ___.	A cart is in equilibrium with 7 N right. If there is only one other horizontal force, what must it be? I can tell because ___.
Example or sketch	**Non-example and why**
Two cables supply 3 N and 5 N upward on a stationary object. What downward force completes the diagram? My example shows ___.	A partner adds another upward 8 N. What would happen to the vertical force sum? I would change ___ because ___.

Instantaneous rest

ES: reposo instantáneo / PT: repouso instantâneo

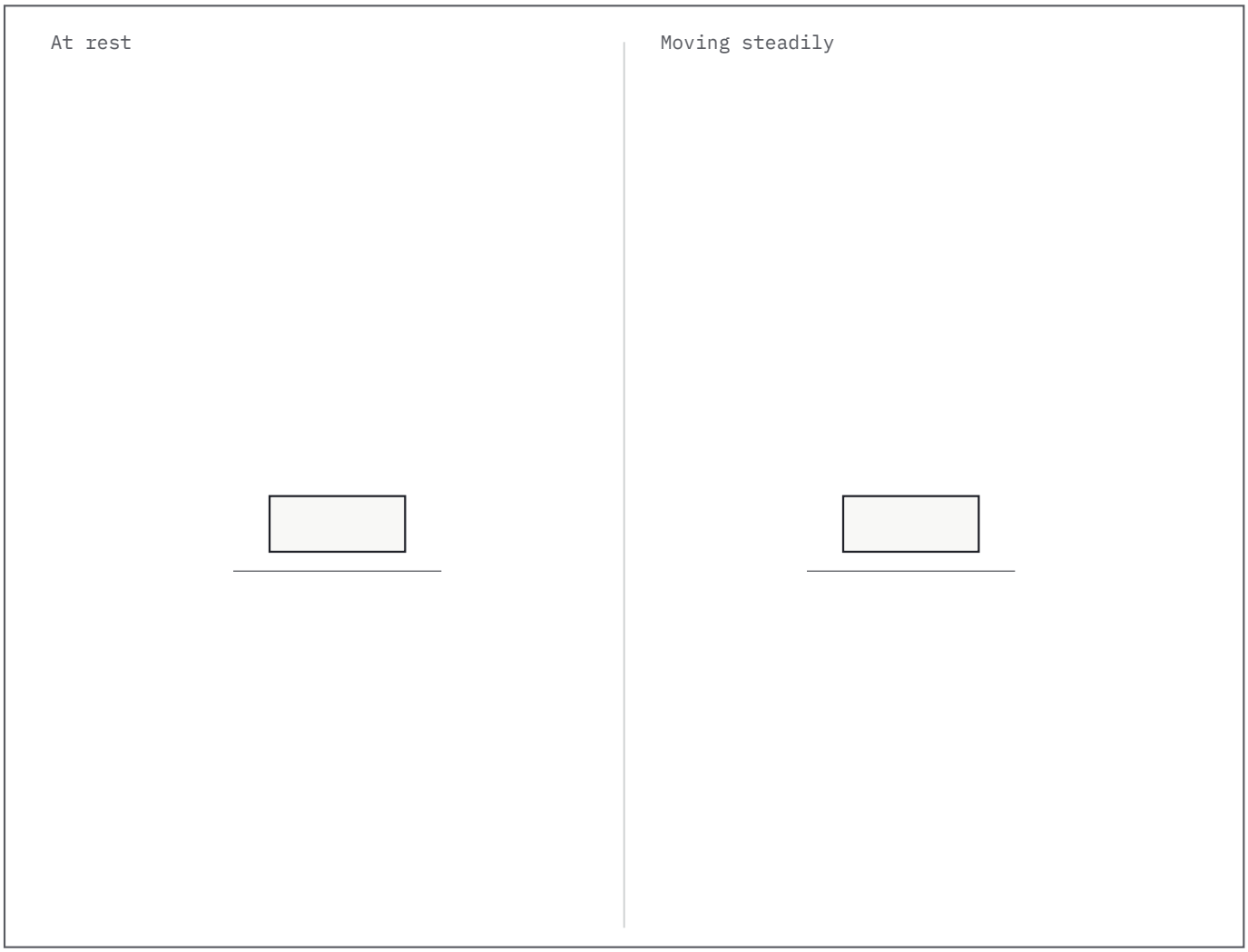
Meaning in my words	How can you tell?
How would you explain instantaneous rest to a friend? In my words, this means ___.	At the top of a ball's straight upward toss, what is zero and what is still acting? I can tell because ___.
Example or sketch	**Non-example and why**
How would you draw velocity and acceleration at that instant? My example shows ___.	"It stopped, so the forces must balance." What does the next instant show? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch two systems with balanced forces: one at rest and one moving steadily. Label the common acceleration.

Labels you can use: balance · upward part · weight.



Tell a partner

The ___ forces balance. Its acceleration is ___.

Stopped or coasting?

18

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Distinguish zero acceleration from zero velocity.

CHOOSE ONE JOB TOGETHER

FLIGHT CONTROLLER ☐

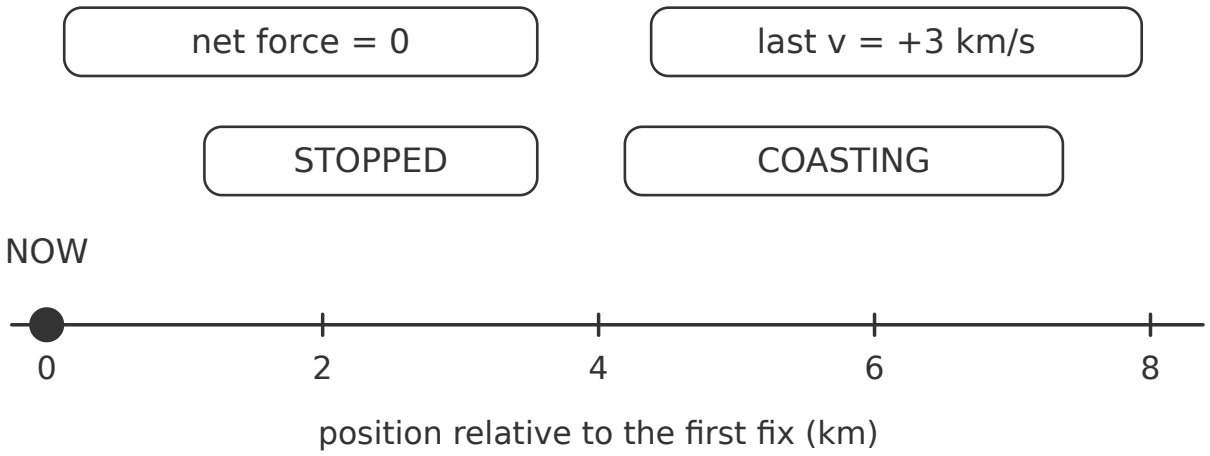
Choose the status graphic. Mark where XJ should be two seconds later relative to the first sighting.

BROADCAST PRODUCER ☐

Choose the on-air label. Explain why zero acceleration and zero speed do not mean the same thing.

Word help: balanced / coasting / acceleration

The flight display reads "net force zero." XJ's last velocity was **+3 km/s**. Two graphics say STOPPED and COASTING.



01 OUR FIRST CALL

Which status should go live? Use a second position to support your call.

The status is ___ because net force ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Can the old status describe the next interval? Explain what the new report changes.

Now we would say ___ because ___.

Switch speakers. Does zero net force mean zero speed?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks